Broadband RF Self-Interference Cancellation for Sub-6 GHz Full-Duplex Radios Using Miniaturized DGS Components

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***Abstract*—In-band full-duplex (IBFD) radios can theoretically double spectral efficiency by enabling simultaneous transmission and reception in the same frequency channel. Their practical realization, however, is limited by strong self-interference (SI) leaking from the local transmitter into the receiver front-end. This paper presents an adaptive analog RF self-interference cancellation (RF-SIC) architecture for the 3.5 GHz sub-6 GHz 5G NR band, with emphasis on wideband operation, miniaturized passive implementation, and robustness against practical hardware impairments. A Defected Ground Structure (DGS)-enhanced passive distribution network is synthesized on RO4350B substrate to reduce footprint while preserving impedance matching, coupling accuracy, and group-delay behavior. The proposed architecture combines electromagnetic/circuit co-design, closed-loop amplitude–phase adaptation, and nonlinear validation under two-tone excitation. Representative co-simulation results indicate a peak RF cancellation depth of 71 dB at 3.5 GHz and an active cancellation bandwidth of 1.0 GHz for a 20 dB-improvement threshold beyond the passive baseline. The design also demonstrates flat group delay over the useful band, mitigation of in-band IMD3 leakage, and analytically bounded degradation in cascaded receiver noise figure due to the injection combiner. These results support the feasibility of compact and broadband RF-SIC front-ends for next-generation full-duplex small-cell and integrated radio platforms.**

***Index Terms*—Defected Ground Structure (DGS), full-duplex, group delay, intermodulation distortion, noise figure, RF self-interference cancellation (RF-SIC), 5G NR.**

I. Introduction

In-band full-duplex (IBFD) wireless systems have attracted strong research interest because they enable simultaneous transmission and reception over the same carrier frequency, thereby potentially doubling link spectral efficiency. In practice, the key obstacle is self-interference (SI), namely the large transmitter leakage coupled into the receiver through antenna reflection, finite circulator/directivity isolation, substrate coupling, and environmental multipath. If not sufficiently suppressed before the receiver low-noise amplifier (LNA) and analog-to-digital converter (ADC), SI may compress the receiver chain, elevate the effective noise floor, and degrade demodulation quality.

Although digital cancellation can remove a large fraction of the residual SI after downconversion, analog RF-domain cancellation remains a mandatory first stage in realistic front-ends. Existing RF-SIC techniques typically trade off cancellation depth, bandwidth, insertion loss, implementation area, and adaptation complexity. Broadband operation is particularly difficult because the cancellation path must reproduce not only the amplitude and phase of the leakage signal at a single tone, but also its dispersive behavior across the occupied channel bandwidth. This requirement becomes more stringent for OFDM-based 5G waveforms, where group delay mismatch directly increases residual error and degrades error vector magnitude (EVM).

This work proposes a compact adaptive RF-SIC architecture centered at 3.5 GHz and implemented with miniaturized DGS-based passive microwave components. Unlike conventional approaches that prioritize cancellation notch depth alone, the proposed design jointly targets footprint reduction, broadband group-delay flatness, and robustness under amplitude/phase tuning errors and nonlinear power amplifier (PA) distortion. The cancellation path is realized using a DGS-enhanced branch-line coupler for compact signal sampling and a Wilkinson combiner for injection at the receiver input, while voltage-controlled attenuation and phase tuning are adjusted in a closed loop.

The main contributions of this paper are as follows:

- A DGS-enhanced passive distribution network for 3.5 GHz RF-SIC that achieves substantial physical miniaturization while maintaining $50\,\Omega$ matching, acceptable coupling balance, and wideband phase behavior.
- A broadband adaptive cancellation topology that achieves deep narrowband suppression and a wide effective cancellation window through joint amplitude–phase optimization and group-delay-aware design.
- A nonlinear and system-level validation framework including IMD3 suppression, cascaded noise figure analysis, tuning sensitivity, and fabrication tolerance discussion relevant to practical sub-6 GHz full-duplex radios.

The remainder of this paper is organized as follows. Section II introduces the RF-SIC mathematical framework and tuning sensitivity. Section III presents the system architecture and design methodology. Section IV discusses the DGS miniaturized passive components. Section V reports linear, nonlinear, and robustness results. Section VI compares the proposed work with representative prior art, and Section VII concludes the paper.

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II. Theory of RF Self-Interference Cancellation

A. Signal Model

Let the self-interference leakage signal arriving at the receiver input be represented by

$$V_{\text{leak}}(\omega) = A_{\text{leak}}(\omega) e^{j\phi_{\text{leak}}(\omega)}. \tag{1}$$

An RF sample of the transmit signal is processed through a tunable cancellation path and reinjected before the LNA:

$$V_{\text{can}}(\omega) = A_{\text{can}}(\omega) e^{j\phi_{\text{can}}(\omega)}. \tag{2}$$

The residual signal is

$$V_{\text{res}}(\omega) = V_{\text{leak}}(\omega) - V_{\text{can}}(\omega). \tag{3}$$

Perfect cancellation requires

$$A_{\text{can}}(\omega) = A_{\text{leak}}(\omega), \qquad \phi_{\text{can}}(\omega) = \phi_{\text{leak}}(\omega). \tag{4}$$

In practice, finite tuner resolution, passive imbalance, and multipath dynamics prevent exact matching across frequency.

B. Cancellation Depth Under Amplitude and Phase Errors

If the cancellation path exhibits amplitude mismatch ϵ dB and phase mismatch $\Delta\theta$, then the achievable cancellation depth can be written as

$$C_{\text{dB}} = -10 \log_{10} \left(1 + 10^{\epsilon/10} - 2 \cdot 10^{\epsilon/20} \cos(\Delta\theta) \right). \tag{5}$$

Equation (5) highlights the stringent sensitivity of deep cancellation to small tuning errors. In the high-suppression regime, sub-degree phase alignment and sub-dB amplitude matching are required. This motivates a closed-loop adaptive controller rather than a one-time open-loop calibration.

C. Broadband Requirement and Group Delay

For wideband OFDM signals, cancellation at a single continuous-wave tone is insufficient. The frequency-dependent phase slope of each path must also be matched. Group delay is given by

$$\tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega}. \tag{6}$$

A mismatch

$$\Delta\tau_g(\omega) = \tau_{g,\text{leak}}(\omega) - \tau_{g,\text{can}}(\omega) \tag{7}$$

causes frequency-selective residual SI, which appears as constellation spreading and EVM degradation after OFDM demodulation. Therefore, the passive structures in the cancellation path must be designed not only for matching and coupling, but also for low group-delay ripple across the intended channel span.

D. Noise Figure Trade-Off

The reinjection of the cancellation signal requires a power combiner placed ahead of the LNA. For the cascaded system,

$$F_{\text{sys}} = F_{\text{comb}} + \frac{F_{\text{LNA}} - 1}{G_{\text{comb}}}, \tag{8}$$

where F_{comb} is the combiner noise factor and $G_{\text{comb}} < 1$ is its gain. Because a passive combiner introduces loss, it directly increases the receiver noise figure. This penalty is accepted to prevent severe LNA desensitization or compression from large SI leakage levels.

Placeholder for system block diagram:
TX $\rightarrow$ sampler/BLC $\rightarrow$ VVA $\rightarrow$ VPS/delay path $\rightarrow$ combiner $\rightarrow$ RX LNA,
with DSP feedback control from receiver observation.

Fig. 1. Proposed adaptive RF self-interference cancellation architecture.

E. Nonlinear Distortion Considerations

In realistic transmitters, the PA generates nonlinear distortion including third-order intermodulation (IMD3). For a two-tone excitation at f_1 and f_2, the in-band distortion terms occur at

$$f_{\text{IMD3,low}} = 2f_1 - f_2, \qquad f_{\text{IMD3,high}} = 2f_2 - f_1. \tag{9}$$

Since these products fall close to the desired receive band, an RF canceller designed only for the main carrier may leave nonlinear spectral regrowth insufficiently suppressed. Consequently, bandwidth and linearity validation must include distorted SI waveforms and not only single-tone notch depth.

III. System Architecture and Design Methodology

A. RF-SIC Architecture

Fig. 1 illustrates the proposed architecture. A coupled sample of the transmit signal is routed through a tunable attenuation and phase control path and then injected into the receiver path through a Wilkinson combiner. The receiver front-end sees the superposition of the direct leakage path and the synthesized cancellation signal. A digital control loop estimates residual SI and updates the tunable elements iteratively.

The overall SI suppression is conceptually divided into passive isolation and active RF cancellation:

$$C_{\text{total}} \approx C_{\text{passive}} + C_{\text{active}}, \tag{10}$$

when expressed in dB under sufficiently independent suppression mechanisms. In the present work, the emphasis is placed on the active RF domain after a baseline passive leakage isolation stage.

B. Substrate and Implementation Platform

The circuit is implemented on Rogers RO4350B laminate with nominal thickness of 20 mil. This material is widely used in RF and microwave PCB implementations because of its stable dielectric constant and low loss tangent over GHz frequencies. For design consistency, the substrate parameters were taken as relative permittivity $\epsilon_r = 3.48$ and low dielectric loss suitable for sub-6 GHz microwave structures. Copper thickness, etch bias, and fabrication tolerances are included in post-layout sensitivity analysis.

C. Closed-Loop Adaptation Strategy

The adaptive controller minimizes the residual SI power measured at the receiver observation node. Let the tuning vector be

$$\mathbf{x} = [a, \phi]^T, \tag{11}$$

Placeholder for DGS branch-line coupler layout: top metal, bottom DGS pattern, and key geometric parameters.

Fig. 2. Geometry of the proposed DGS-enhanced branch-line coupler.

where a is attenuation control and ϕ is phase control. The optimization target is

$$J(\mathbf{x}) = \int_{\omega_1}^{\omega_2} |V_{\text{res}}(\omega; \mathbf{x})|^2 \, d\omega. \tag{12}$$

In implementation, this cost can be evaluated either at a single tone for narrowband calibration or over multiple frequency bins for wideband adaptation. A derivative-free search such as Nelder–Mead is attractive because it avoids explicit gradient estimation in the presence of measurement noise and tuner non-idealities. The algorithm iteratively updates the control voltages of the VVA and variable phase shifter (VPS) until the reduction in residual SI falls below a predefined threshold.

D. EM/Circuit Co-Simulation Flow

The passive microwave blocks were first synthesized using transmission-line theory and then refined with 3-D electromagnetic simulation. Their S-parameter models were imported into ADS and combined with the tunable attenuation/phase elements for full system co-simulation. Harmonic balance analysis was subsequently used to evaluate two-tone nonlinear cancellation performance under PA-induced IMD3.

IV. Miniaturized DGS Passive Network

A. DGS-Enhanced Branch-Line Coupler

A conventional quarter-wave branch-line coupler at 3.5 GHz occupies significant area on standard microwave substrate. To reduce size, DGS cells were etched beneath selected microstrip sections. The perturbation of the ground-return current path introduces effective distributed inductive and capacitive loading, which increases the slow-wave factor and shortens the physical length needed to realize a given electrical phase shift.

The equivalent transmission-line section can be interpreted through an augmented LC model:

$$Z_{\text{eff}}(\omega) \approx j\omega L_s + \frac{1}{j\omega C_p}, \tag{13}$$

where the effective series inductance L_s and shunt capacitance C_p are controlled by the DGS geometry. This mechanism enables miniaturization while preserving the required quarter-wave behavior near the design frequency.

For the chosen substrate and center frequency, the conventional branch-line coupler occupies approximately $252{,}004\ \text{mil}^2$, whereas the proposed DGS implementation occupies approximately $144{,}400\ \text{mil}^2$. The corresponding miniaturization is

$$\eta_{\text{mini}} = \frac{A_{\text{conv}} - A_{\text{DGS}}}{A_{\text{conv}}} \times 100\% \approx 42.7\%. \tag{14}$$

The key design goal is not area reduction alone, but area reduction without unacceptable degradation in return loss, coupling balance, isolation, or group-delay ripple.

Placeholder for passive block S-parameters: S_{11}, coupling, insertion loss, isolation, imbalance.

Fig. 3. Representative S-parameters of the DGS coupler and Wilkinson combiner.

TABLE I
Passive Block Characterization Metrics

Metric	DGS Coupler	Wilkinson
Center frequency	3.5 GHz	3.5 GHz
Return loss at f_0	[add]	[add]
Coupling / split ratio	[add]	[add]
Isolation at f_0	[add]	[add]
Amplitude imbalance	[add]	[add]
Phase imbalance	[add]	[add]
Insertion loss	[add]	[add]
Group delay ripple	[add]	[add]

B. Wilkinson Combiner

A symmetric Wilkinson power combiner is used at the receiver injection point because it provides good port matching and high isolation between the through and cancellation paths. The nominal characteristic impedance and isolation resistor values were selected according to standard equal-split design equations. The isolation resistor dissipates imbalance energy during adaptation and prevents excessive interaction between the ports when the cancellation path is temporarily mistuned.

C. Component Characterization Metrics

To fully validate the passive network, the following metrics are extracted for both the DGS coupler and Wilkinson combiner over frequency:

- Return loss and port matching (S_{11}, S_{22}, S_{33}),
- Coupling or division ratio (S_{21}, S_{31}),
- Isolation between coupled ports,
- Amplitude imbalance and phase imbalance,
- Insertion loss,
- Group delay and group-delay ripple.

V. Simulation Results and Discussion

A. Passive Block Performance

Fig. 3 should present the S-parameter results of the DGS coupler and Wilkinson combiner. In addition to center-frequency response, the frequency dependence of matching, coupling, and isolation must be reported to demonstrate that the size reduction does not come at the expense of unstable broadband behavior.

Table I summarizes the recommended metrics to report after EM optimization.

B. Linear RF Cancellation

The linear co-simulation combines the passive structures with tunable attenuation and phase control. After optimization, the cancellation path is set to approximately 23.7 dB attenuation and 270° phase shift, yielding a peak cancellation depth of 71 dB at 3.5 GHz. The baseline passive isolation is approximately 34 dB, and the active cancellation bandwidth

Placeholder for RF cancellation response:
baseline leakage, tuned cancellation, and residual SI vs frequency.

Fig. 4. Simulated RF cancellation response across frequency.

Placeholder for sensitivity plots:
(a) attenuation sweep,
(b) phase sweep,
(c) contour of cancellation depth from Eq. (1).

Fig. 5. Sensitivity of cancellation depth to amplitude and phase mismatch.

Placeholder for group delay plot:
$\tau_{g,\text{leak}}$, $\tau_{g,\text{can}}$, and mismatch vs frequency.

Fig. 6. Group delay behavior of leakage and cancellation paths.

is defined here as the frequency span over which the total suppression exceeds the baseline by at least 20 dB.

Under this definition, the proposed design exhibits a 1.0 GHz active cancellation window from 3.3 GHz to 4.3 GHz. Because bandwidth definitions vary significantly across the literature, explicitly stating the threshold used for cancellation bandwidth is essential for fair comparison.

C. Sensitivity to Tuning Error

To verify the theory of (5), parameter sweeps are performed around the optimum control point. Amplitude perturbation primarily reduces notch depth while phase perturbation alters both notch depth and notch frequency alignment. A 2-D contour plot of predicted cancellation depth versus amplitude and phase error is recommended to show the narrow region where > 40 dB suppression is sustained.

D. Group Delay and Wideband Suitability

Fig. 6 should report the group delay of the cancellation path, leakage path, and their mismatch. Outside the singular behavior near the exact notch frequency, the designed path maintains approximately flat group delay around 0.4 ns over the useful region. This supports broadband cancellation of wideband OFDM signals and reduces frequency-selective residual SI that would otherwise degrade EVM.

E. Nonlinear Two-Tone Validation

A two-tone harmonic balance test is used to examine whether the canceller also suppresses nonlinear in-band distortion. The excitation tones are set to $f_1 = 3.50$ GHz and $f_2 = 3.51$ GHz with a representative transmit power level of +10 dBm at the PA output model input. In the uncompensated case, the receiver sees both the fundamental leakage and adjacent IMD3 terms. After tuning, the fundamental SI is strongly reduced and the IMD3 products are also suppressed toward the simulated noise floor, demonstrating that the cancellation bandwidth is sufficiently broad to mitigate nearby spectral regrowth.

Placeholder for harmonic balance spectrum:
before/after RF cancellation around 3.5 GHz with IMD3 markers.

Fig. 7. Two-tone harmonic balance spectrum showing suppression of fundamental SI and IMD3 products.

TABLE II
EXAMPLE SELF-INTERFERENCE AND RECEIVER BUDGET

Quantity	**Value**
TX leakage at receiver without SIC	[add] dBm
Passive isolation only	[add] dB
Additional active RF cancellation	[add] dB
Residual SI at LNA input	[add] dBm
Combiner insertion loss	[add] dB
LNA gain	[add] dB
LNA noise figure	[add] dB
Estimated cascaded NF	[add] dB

F. Noise Figure and Receiver Budget

Table II is recommended to quantify the receiver protection benefit against the penalty of the combiner loss. Even though the Wilkinson introduces a noise figure penalty, this trade-off is acceptable if it prevents the LNA from entering compression due to large residual SI.

G. Environmental and Fabrication Robustness

In practical deployments, SI is not static. Nearby reflectors, antenna movement, housing effects, and mismatch changes alter the leakage channel over time. Therefore, the paper should include at least one robustness experiment or simulation where the leakage path phase and magnitude are perturbed to emulate environmental change. The corresponding retuned and unretuned cancellation responses will demonstrate the value of closed-loop adaptation.

Manufacturing variation must also be considered. Parameters such as dielectric constant tolerance, etch-width variation, copper thickness, and resistor tolerance can perturb the passive structures and shift the optimum tuning point. A Monte Carlo or worst-case sweep should therefore be included to quantify the distribution of notch depth, return loss, and group-delay ripple under fabrication uncertainty.

VI. Comparison With Prior Art

Table III compares the proposed work with representative RF-SIC implementations. To make the comparison fair, bandwidth should always be defined with its suppression threshold, and the table should distinguish passive isolation from active RF cancellation whenever the cited papers report them separately.

VII. Discussion

The results indicate that DGS miniaturization can be integrated into RF-SIC front-ends without inherently sacrificing broadband cancellation behavior, provided that group-delay flatness and passive imbalance are explicitly included in the design objectives. This is important because size reduction

TABLE III
COMPARISON WITH REPRESENTATIVE SUB-6 GHZ RF-SIC ARCHITECTURES

Work	Freq.	Peak SIC	BW	BW Definition	TX Power	Adaptive	Nonlinear Eval.	Size
Ref. [1]	2.4 GHz	[add]	[add]	[add]	[add]	Yes/No	Yes/No	Standard
Ref. [2]	3.5 GHz	[add]	[add]	[add]	[add]	Yes/No	Yes/No	Standard
Ref. [3]	[add]	[add]	[add]	[add]	[add]	Yes/No	Yes/No	[add]
This work	3.5 GHz	71 dB	1.0 GHz	$>$ 20 dB over baseline	+10 dBm†	Yes	Yes	Miniaturized

†Representative nonlinear test condition used in the present study.

alone is insufficient for full-duplex radios; the cancellation path must remain spectrally faithful to the SI leakage over the occupied channel.

The proposed architecture also highlights the central system trade-off of RF-SIC design. Deep cancellation improves receiver linearity margin and protects the LNA/ADC chain, but the feed-in combiner increases front-end loss and therefore worsens the cascaded noise figure. The design target is thus not maximum notch depth in isolation, but an acceptable compromise among cancellation depth, bandwidth, adaptation speed, hardware complexity, and receiver sensitivity.

Several limitations should be noted. First, the presented validation is based primarily on simulation and co-simulation; measured hardware results remain necessary for publication-quality confirmation. Second, PA memory effects and higher-order nonlinearities may require richer behavioral models. Third, practical full-duplex platforms generally combine passive isolation, analog RF cancellation, and digital baseband cancellation, so the proposed RF block should ultimately be evaluated as part of a multi-stage SI suppression chain.

VIII. CONCLUSION

A broadband and compact RF self-interference cancellation architecture for sub-6 GHz full-duplex radios has been presented. By combining DGS-enhanced passive miniaturization with adaptive amplitude–phase tuning, the proposed design achieves strong cancellation depth, broad effective bandwidth, and controlled group-delay behavior suitable for wideband 5G-like signals. Nonlinear and system-level analyses further show that the architecture can suppress IMD3-contaminated SI while maintaining a manageable receiver noise figure penalty. These characteristics make the proposed approach a promising candidate for compact full-duplex front-ends in small-cell and integrated wireless systems.

Future work will focus on fabricated prototype measurement, over-the-air validation under dynamic multipath conditions, and joint optimization with digital SIC for end-to-end EVM and throughput improvement.

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